

Configure and Test Basic Network Connectivity

Cisco Packet Tracer Lab Project Documentation

1. Project Overview

This Cisco Packet Tracer project covers the foundational configuration tasks required to set up and verify end-to-end network connectivity in a simulated LAN environment. Spanning six lab videos, the project progressively builds skills from basic switch configuration all the way through dual-pool DHCP service for both wired and wireless clients.

Topics covered across all labs:

- Basic switch configuration and configuration persistence (running-config vs. startup-config)
- Static IP addressing for end devices and router interfaces
- Switch management using a VLAN 1 Switch Virtual Interface (SVI)
- Telnet remote access via VTY lines
- Connectivity testing and fault isolation using ping
- DHCP pool configuration on a Cisco IOS router for wired hosts
- DHCP pool configuration for wireless clients with excluded addresses
- Validation of configuration persistence and end-to-end communication

2. Lab Environment

Tool / Component	Details
Simulation Software	Cisco Packet Tracer
Switch	Cisco 2960-24TT (Layer 2 switch)
Router	Cisco ISR 4331 (gateway-router / WAN-router)
End Devices	PC, Laptop, Server-PT (HTTP), Tablet PC, Printer, Smartphone
Wireless Infrastructure	Access Point (Video 7)
Configuration Methods	CLI (Cisco IOS), Packet Tracer Desktop/Config GUI

3. Skills Demonstrated

- Cisco IOS CLI navigation (user EXEC → privileged EXEC → global configuration modes)
- Switch hostname and enable secret configuration
- Understanding running-config vs. startup-config and configuration persistence
- Saving configuration with copy running-config startup-config
- Router interface IP address assignment and activation (no shutdown)
- Static IP addressing for PCs, laptops, and servers
- VLAN 1 Switch Virtual Interface (SVI) configuration for Layer 2 switch management
- VTY line configuration for Telnet remote access
- DHCP pool configuration on a Cisco IOS router (network, default-router, excluded addresses)
- Dual-pool DHCP for wired (192.168.1.0/24) and wireless (192.168.2.0/24) networks
- Connectivity testing and physical layer fault isolation using ping
- Verification commands: show running-config, show startup-config, show ip interface brief, show ip dhcp pool, show vlan

4. IP Addressing Summary

Device	Interface	IP Address	Subnet Mask	Method
Office PC	FastEthernet0	192.168.1.10	255.255.255.0	Static / DHCP
Bob's Laptop	FastEthernet0	192.168.1.20	255.255.255.0	Static / DHCP
HTTP Server	FastEthernet0	192.168.1.30	255.255.255.0	Static / DHCP
main-switch (SVI)	VLAN 1	192.168.1.40	255.255.255.0	Static
gateway-router	GigabitEthernet0/0/1	192.168.1.1	255.255.255.0	Static
Wireless Clients	Wireless NIC	192.168.2.x (pool)	255.255.255.0	DHCP
WAN Router (wireless GW)	LAN Interface	192.168.2.1	255.255.255.0	Static

1. Basic Switch Configuration and Configuration Persistence

A. Objective

Learn how to open a switch CLI in Packet Tracer, apply basic configuration (hostname and enable secret), understand the difference between running-config and startup-config, and verify that saving configuration is required for it to survive a reload.

B. Step-by-Step Procedure

1. In Packet Tracer, click on the 2960-24TT switch, select the CLI tab, and press any key to begin.
2. Enter privileged EXEC mode.
3. Enter global configuration mode.
4. Set the switch hostname to cisco-switch.
5. Configure an enable secret password of 'cisco'.
6. Return to privileged EXEC mode using Ctrl+Z.
7. Type disable to drop back to user EXEC mode, then type enable and enter the password 'cisco' to verify the enable secret is working.
8. View the current running configuration.
9. Initiate a reload of the switch.
10. When prompted "Save? [yes/no]:", type no and press Enter. Confirm the reload.
11. After the switch restarts, notice the hostname reverts to Switch — this confirms the config was NOT saved.
12. Run show running-config and show startup-config to compare both files.
13. Re-enter global configuration mode and reapply the hostname and enable secret.
14. Exit configuration mode and save the configuration to NVRAM.
15. Reload the switch again (confirm when prompted).
16. After restart, verify that the hostname is still cisco-switch and the enable secret is still active — this confirms the config WAS saved.

C. CLI Commands

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname cisco-switch
cisco-switch(config)# enable secret cisco
cisco-switch(config)# end

! --- Verify the enable secret works ---
cisco-switch# disable
cisco-switch> enable
Password: cisco

! --- View the running configuration ---
cisco-switch# show running-config

! --- Reload WITHOUT saving ---
cisco-switch# reload
  System configuration has been modified. Save? [yes/no]: no
  Proceed with reload? [confirm] <Enter>

! --- After reload: config is gone ---
Switch# show running-config
Switch# show startup-config

! --- Reconfigure and SAVE ---
Switch# configure terminal
Switch(config)# hostname cisco-switch
cisco-switch(config)# enable secret cisco
cisco-switch(config)# exit
cisco-switch# copy running-config startup-config
  Destination filename [startup-config]? <Enter>

! --- Reload WITH config saved ---
cisco-switch# reload

! --- After reload: verify config persists ---
cisco-switch# enable
Password: cisco
cisco-switch# show running-config
cisco-switch# show startup-config
```

D. Command Explanations

Command	Purpose
enable	Enters privileged EXEC mode from user EXEC mode. Required before any configuration or show commands.
configure terminal	Enters global configuration mode. Raw notes used the shorthand 'confi t' — the full command is configure terminal.
hostname cisco-switch	Sets the device's hostname. This label appears in the CLI prompt and identifies the device in documentation.
enable secret cisco	Sets an encrypted privileged EXEC password. 'enable secret' uses MD5 encryption and is stronger than 'enable password'.

<code>end / Ctrl+Z</code>	Exits any configuration submode and returns directly to privileged EXEC mode.
<code>show running-config</code>	Displays the current configuration held in RAM. Raw notes used the shorthand 'sh run'.
<code>reload</code>	Restarts the device. If config is unsaved, it is lost on restart.
<code>copy running-config startup-config</code>	Saves the RAM-based running config to NVRAM startup-config. Raw notes used the shorthand 'copy run start'.
<code>show startup-config</code>	Displays the NVRAM-saved configuration that loads on boot. Raw notes used 'sh start'.

E. Verification / Expected Results

- After the first reload (without saving): the CLI prompt should show Switch# — the hostname 'cisco-switch' is gone.
- After copy running-config startup-config and a second reload: the prompt immediately shows cisco-switch# after restart.
- show startup-config should display the hostname and enable secret settings that were saved.

F. Key Learning Point

Running-Config vs. Startup-Config

Running-config lives in volatile RAM and is active immediately but is erased on power-off or reload unless saved. Startup-config lives in non-volatile NVRAM and is loaded at boot. Always run 'copy running-config startup-config' before reloading or powering down a device.

2. Configure Static IP Addresses and Router Interface

A. Objective

Assign static IPv4 addresses to three end devices (PC, Laptop, Server) and configure the connected router interface with its gateway IP address. Verify the interface status using show commands.

B. Step-by-Step Procedure

17. In Packet Tracer, go to File > Open > Labs and open the 'config ip add' lab file.
18. Click on the Office PC > Config tab > FastEthernet0. Set the IP configuration to Static. Enter IP: 192.168.1.10, Subnet Mask: 255.255.255.0.
19. Click on Bob's Laptop > Config tab > FastEthernet0. Set to Static. Enter IP: 192.168.1.20, Subnet Mask: 255.255.255.0.
20. Click on the HTTP Server > Config tab > FastEthernet0. Set to Static. Enter IP: 192.168.1.30, Subnet Mask: 255.255.255.0.
21. Click on the gateway-router (ISR4331) > CLI tab. Enter privileged EXEC mode and then global configuration mode.
22. Enter the GigabitEthernet0/0/1 interface configuration mode.
23. Assign the IP address 192.168.1.1 with subnet mask 255.255.255.0.
24. Activate the interface with no shutdown.
25. Return to privileged EXEC mode and save the configuration.
26. Verify the interface is up using show ip interface brief.
27. Verify interface details using show ip interface GigabitEthernet0/0/1.

C. CLI Commands

```
! --- Router interface configuration ---
Router> enable
Router# configure terminal
Router(config)# interface GigabitEthernet0/0/1
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# end

! --- Save configuration ---
Router# copy running-config startup-config
Destination filename [startup-config]? <Enter>

! --- Verify interface status ---
Router# show ip interface brief
Router# show ip interface GigabitEthernet0/0/1
```

D. Command Explanations

- `interface GigabitEthernet0/0/1` — Enters configuration mode for the specified router interface. On the ISR4331, LAN ports use the `GigabitEthernet0/0/x` notation.
- `ip address 192.168.1.1 255.255.255.0` — Assigns the Layer 3 IP address and subnet mask to the interface. This becomes the default gateway for all hosts in the 192.168.1.0/24 network.
- `no shutdown` — Activates the interface. Router interfaces are administratively disabled by default and must be explicitly enabled.
- `show ip interface brief` — Displays a condensed table of all interfaces with their IP addresses and status (up/up, up/down, administratively down). The target interface should show 'up' in both columns.
- `show ip interface GigabitEthernet0/0/1` — Displays full Layer 3 details for a specific interface, including IP address, MTU, and helper addresses.

E. Verification / Expected Results

- `show ip interface brief` should list `GigabitEthernet0/0/1` with IP 192.168.1.1 and status 'up / up'.
- All three end devices should show their assigned static IPs in the Config tab.
- A ping from any device to 192.168.1.1 (the router) should succeed.

F. Key Learning Point

Default Gateway

The router's interface IP (192.168.1.1) serves as the default gateway for all devices on the 192.168.1.0/24 subnet. Without a gateway configured on end devices, traffic cannot leave the local subnet. Router interfaces are shutdown by default — 'no shutdown' is mandatory.

3. Configure Switch Virtual Interface and Telnet Access

A. Objective

Configure a management IP address on the Layer 2 switch using a Switch Virtual Interface (SVI) on VLAN 1. Configure VTY lines to allow Telnet remote access to the switch. Demonstrate Telnet login from a laptop to both the router and the switch.

B. Step-by-Step Procedure

28. Open the 'virtual interfaces' lab file in Packet Tracer.
29. Test connectivity first: from Bob's Laptop > Desktop > Telnet/SSH Client, enter the router's IP (192.168.1.1) and connect using the password 'cisco'. This confirms the router is reachable.
30. On the main-switch CLI, enter privileged EXEC mode and global configuration mode.
31. Configure the VLAN 1 SVI with the management IP address.
32. Activate the VLAN 1 interface with no shutdown.
33. Return to privileged EXEC mode and verify VLAN status with show vlan.
34. Configure the VTY lines 0 through 5 for Telnet access.
35. Set the VTY Telnet password.
36. Enable password checking on VTY lines with the login command.
37. Restrict remote access to Telnet only.
38. Configure the enable secret so that after Telnet login, privileged access is also protected.
39. Exit and save the configuration.
40. From Bob's Laptop > Desktop > Telnet/SSH Client, Telnet to 192.168.1.40. Enter the VTY password 'cisco', then type enable and enter the enable secret 'cisco' to reach privileged mode.

C. CLI Commands

```
! === SWITCH MANAGEMENT IP (SVI) ===
Switch> enable
Switch# configure terminal

! Correct: assign IP to VLAN 1 SVI (not a physical port)
Switch(config)# interface vlan 1
Switch(config-if)# ip address 192.168.1.40 255.255.255.0
Switch(config-if)# no shutdown
Switch(config-if)# end

! Verify VLAN status
Switch# show vlan

! === VTY / TELNET CONFIGURATION ===
Switch# configure terminal
Switch(config)# line vty 0 5
Switch(config-line)# password cisco
Switch(config-line)# login
Switch(config-line)# transport input telnet
Switch(config-line)# exit

! Protect privileged EXEC access after Telnet login
```



```
Switch(config)# enable secret cisco
Switch(config)# end

! Save configuration
Switch# copy running-config startup-config
  Destination filename [startup-config]? <Enter>

! === FROM LAPTOP: TELNET TO SWITCH ===
! Laptop > Desktop > Telnet/SSH Client
! Host: 192.168.1.40 > Connect
! Password: cisco
! Switch> enable
! Password: cisco
! Switch#
```

D. Command Explanations

- interface vlan 1 — Enters the SVI for VLAN 1. This is the correct management interface on a Layer 2 Cisco switch. All switch ports are in VLAN 1 by default, so this IP is reachable from any connected device.
- ip address 192.168.1.40 255.255.255.0 — Assigns the switch management IP. Administrators use this address to Telnet or SSH into the switch remotely.
- no shutdown — Activates the SVI. VLAN interfaces may be administratively down if the VLAN itself has no active ports.
- line vty 0 5 — Configures virtual terminal lines 0 through 5 (6 simultaneous remote sessions). VTY lines are used for Telnet and SSH access.
- password cisco — Sets the password required when connecting via Telnet.
- login — Tells the switch to require a password when a user connects to the VTY line. Without this command, remote users are not prompted for a password.
- transport input telnet — Restricts VTY access to Telnet only. For production environments, SSH (transport input ssh) is preferred for security.
- enable secret cisco — Encrypts the privileged EXEC password. After Telnetting in, a user must still type 'enable' and provide this secret to reach privileged mode.

E. Verification / Expected Results

- show vlan should display VLAN 1 as active with all switch ports assigned to it.
- From the laptop, Telnetting to 192.168.1.40 should prompt for a password. Entering 'cisco' should bring up Switch> user mode. Typing 'enable' and entering 'cisco' should bring up Switch# privileged mode.

F. Key Learning Point

Layer 2 Switch Management: Always Use interface vlan 1

A Cisco 2960 is a Layer 2 switch. Physical switchports do not have IP addresses — they only forward frames. To manage the switch remotely, assign an IP to the VLAN 1 SVI (interface vlan 1). This virtual interface represents the entire VLAN logically and makes the switch IP-addressable for Telnet/SSH access.

4. Connectivity Testing

A. Objective

Use ping to validate end-to-end Layer 3 connectivity between all configured devices. Simulate a physical link failure by disconnecting the server and observe how ping reveals the fault. Reconnect and confirm recovery.

B. Step-by-Step Procedure

41. Open the 'connectivity' lab file.
42. From Bob's Laptop > Desktop > Command Prompt, ping each device in the network.
43. From the main-switch CLI, ping host addresses to verify the switch can reach devices on its management subnet.
44. Simulate a fault: physically disconnect the cable between the HTTP Server and the main-switch in the Packet Tracer topology.
45. From the laptop command prompt, ping 192.168.1.30 (the server). Observe the failure.
46. From the main-switch CLI, also ping 192.168.1.30. Observe the failure.
47. Reconnect the server's cable to the switch in the topology.
48. Re-run the ping tests. Verify that connectivity to 192.168.1.30 is restored from both the laptop and the switch.

C. CLI Commands / Tests

```
! === FROM LAPTOP COMMAND PROMPT ===
ping 192.168.1.10      ! Office PC
ping 192.168.1.30      ! HTTP Server
ping 192.168.1.1       ! Gateway Router
ping 192.168.1.40      ! main-switch SVI

! === FROM MAIN-SWITCH CLI ===
main-switch# ping 192.168.1.10      ! Office PC
main-switch# ping 192.168.1.30      ! HTTP Server

! --- Disconnect server cable in Packet Tracer ---

! Ping now fails:
main-switch# ping 192.168.1.30
.....
Success rate is 0 percent

! --- Reconnect server cable ---

! Ping succeeds again:
main-switch# ping 192.168.1.30
!!!!!!
Success rate is 100 percent
```

D. Understanding Ping Output

In Cisco IOS ping output, each character represents one ICMP echo request/reply cycle:

- ! — Success: ICMP reply received
- . — Timeout: no reply received within the wait period
- U — Destination unreachable: the host or network returned an ICMP Unreachable message

E. Verification / Expected Results

- Before disconnecting: all four pings from the laptop should return 5 exclamation marks (!!!!!), indicating 100% success.
- After disconnecting the server: pings to 192.168.1.30 should time out (.....) from both the laptop and the switch.
- After reconnecting: pings to 192.168.1.30 resume immediately.

F. Key Learning Point

Physical Connectivity and Fault Isolation

Ping is a Layer 3 tool, but a ping failure can indicate problems at any layer below Layer 3 — including physical link failures. This exercise demonstrates that a disconnected cable results in an immediate ping failure and that restoring the cable restores connectivity. This is a foundational troubleshooting workflow: test, isolate, fix, retest.

5. DHCP Configuration

A. Objective

Configure the gateway router as a DHCP server for the 192.168.1.0/24 wired LAN. Switch end devices from static to dynamic IP assignment and verify that they receive correct addresses from the pool.

B. Step-by-Step Procedure

49. Open the 'DHCP server add' lab file.
50. On the gateway-router CLI, enter privileged EXEC mode (password: cisco) then global configuration mode.
51. Create a DHCP pool named 'dhcp-pool1'.
52. Set the default gateway (router IP) for clients.
53. Set the network address for the DHCP pool.
54. Return to privileged EXEC mode and save the configuration.
55. On each end device (Office PC, Laptop, HTTP Server), navigate to Config > FastEthernet0 and change the IP assignment from Static to DHCP.
56. Verify on the router with show ip dhcp pool and show running-config | section dhcp.
57. Test connectivity by pinging from the router and from the Office PC.

C. CLI Commands

```
! === DHCP POOL CONFIGURATION ON ROUTER ===
Router> enable
Password: cisco
Router# configure terminal

Router(config)# ip dhcp pool dhcp-pool1
Router(dhcp-config)# default-router 192.168.1.1
Router(dhcp-config)# network 192.168.1.0 255.255.255.0
Router(dhcp-config)# end

Router# copy running-config startup-config
Destination filename [startup-config]? <Enter>

! === VERIFY DHCP ===
Router# show ip dhcp pool
Router# show running-config | section dhcp

! === CONNECTIVITY TEST FROM ROUTER ===
Router# ping 192.168.1.1
Router# ping 192.168.1.2
Router# ping 192.168.1.3

! === FROM OFFICE PC COMMAND LINE ===
ping 192.168.1.1
```

D. Command Explanations

- `ip dhcp pool dhcp-pool1` — Creates a DHCP address pool named 'dhcp-pool1' and enters DHCP configuration submode.
- `network 192.168.1.0 255.255.255.0` — Defines the subnet from which IP addresses will be leased to clients. This must be a network address, not a host address.
- `default-router 192.168.1.1` — Tells DHCP clients to use 192.168.1.1 as their default gateway. This is the router's own interface IP.
- `show ip dhcp pool` — Displays all configured DHCP pools and statistics including how many addresses have been leased.
- `show running-config | section dhcp` — Filters the running configuration to display only DHCP-related lines. The pipe (|) with 'section' is a useful IOS output filter.

E. Verification / Expected Results

- After switching PCs to DHCP, each device should receive an IP from the 192.168.1.0/24 pool (the router itself holds .1, so clients will receive .2, .3, .4, etc.).
- `show ip dhcp pool` should show leases have been assigned.
- Pings to 192.168.1.1 from the Office PC should succeed, confirming full connectivity.

F. Key Learning Point

Router as DHCP Server

A Cisco IOS router can act as a DHCP server without any additional software. The 'ip dhcp pool' command creates the pool, 'network' defines the assignable range, and 'default-router' provides the gateway to clients. Successful dynamic IP assignment followed by a working ping confirms the entire DHCP process — discovery, offer, request, and acknowledgment (DORA) — completed correctly.

6. Capstone: DHCP for Wired and Wireless Networks

A. Objective

Configure two separate DHCP pools on the WAN router — one for wireless clients (192.168.2.0/24) and one for wired hosts (192.168.1.0/24). Exclude the gateway address from each pool. Verify that all client types (laptops, tablets, PCs, printer) receive dynamic addresses and can communicate.

B. Step-by-Step Procedure

58. Open the capstone lab file in Packet Tracer. Observe the topology: a WAN router connects a wireless access point segment (192.168.2.0/24) on the left and a wired switch segment (192.168.1.0/24) on the right.
59. On the WAN router CLI, enter privileged EXEC mode and global configuration mode.
60. Create the DHCP pool for the wireless network.
61. Set the network address for wireless clients.
62. Set the default router for wireless clients.
63. Exclude the gateway address from the wireless DHCP pool so the router's own interface IP is never accidentally leased.
64. Create the DHCP pool for wired hosts.
65. Set the network address for wired hosts.
66. Set the default router for wired clients.
67. Exclude the gateway address from the wired DHCP pool.
68. On the wireless side, set User Laptop, Tablet PC, and Admin Laptop to DHCP via their wireless NIC settings.
69. Verify the wireless clients receive addresses in the 192.168.2.x range.
70. Test connectivity: from a wireless client, ping the gateway 192.168.2.1.
71. On the wired side, set User PC, User PC (second), and Printer to DHCP via their Ethernet NIC settings.
72. Verify wired clients receive addresses in the 192.168.1.x range.
73. From PC2, ping 192.168.1.1 (wired gateway) and attempt to ping a wireless host (e.g., 192.168.2.2) to test inter-subnet routing.

C. CLI Commands

```
! === WIRELESS DHCP POOL ===
WAN_Router> enable
WAN_Router# configure terminal

! Exclude gateway IP from wireless pool
WAN_Router(config)# ip dhcp excluded-address 192.168.2.1

! Create wireless DHCP pool
WAN_Router(config)# ip dhcp pool wireless
WAN_Router(dhcp-config)# network 192.168.2.0 255.255.255.0
WAN_Router(dhcp-config)# default-router 192.168.2.1
WAN_Router(dhcp-config)# exit

! === WIRED DHCP POOL ===

! Exclude gateway IP from wired pool
WAN_Router(config)# ip dhcp excluded-address 192.168.1.1

! Create wired DHCP pool
WAN_Router(config)# ip dhcp pool hosts
WAN_Router(dhcp-config)# network 192.168.1.0 255.255.255.0
WAN_Router(dhcp-config)# default-router 192.168.1.1
WAN_Router(dhcp-config)# end

WAN_Router# copy running-config startup-config
Destination filename [startup-config]? <Enter>

! === VERIFY ===
WAN_Router# show ip dhcp pool
WAN_Router# show running-config | section dhcp

! === CONNECTIVITY TESTS ===
! From wireless laptop command prompt:
ping 192.168.2.1

! From wired PC2 command prompt:
ping 192.168.1.1
ping 192.168.2.2      ! Test inter-subnet routing (if route exists)
```

D. Command Explanations

- `ip dhcp excluded-address 192.168.x.1` — Reserves the gateway's own IP so it is never assigned to a DHCP client. This prevents an address conflict. This command is entered in global config mode, before or after the pool definition.
- `ip dhcp pool wireless` — Creates a DHCP pool named 'wireless'. The name is arbitrary but should be descriptive.
- `network 192.168.2.0 255.255.255.0` — Defines the IP address range to be leased to wireless clients. The .0 address is the network address — not a host address.
- `default-router 192.168.2.1` — Tells wireless clients to use 192.168.2.1 as their gateway.
- `ip dhcp pool hosts` — Second pool for the wired 192.168.1.0/24 network.

E. Verification / Expected Results

- Wireless clients (Laptop, Tablet, Admin Laptop) should receive IPs in the 192.168.2.2–254 range.
- Wired clients (User PCs, Printer) should receive IPs in the 192.168.1.2–254 range.
- `show ip dhcp pool` should show two pools: 'wireless' and 'hosts', both with active leases.
- Pings from wireless clients to 192.168.2.1 and from wired clients to 192.168.1.1 should both succeed.
- Cross-subnet pings (192.168.1.x to 192.168.2.x) succeed if the WAN router has routing enabled between the two subnets.

F. Key Learning Point

Dual-Pool DHCP and Excluded Addresses

A single Cisco IOS router can serve multiple DHCP pools — one per subnet. The `'ip dhcp excluded-address'` command is critical: it prevents the router from leasing its own interface IPs to clients, which would cause duplicate IP conflicts. This capstone lab demonstrates a realistic small-office topology with both wired and wireless client segments managed by a single router.

Final Project Summary

Final Project Summary

This six-lab Cisco Packet Tracer project provides a comprehensive walkthrough of foundational Cisco IOS network configuration. Beginning with the fundamentals of switch configuration and configuration persistence, the project progresses through static IP addressing, router interface activation, Layer 2 switch management via SVI, Telnet remote access, and DHCP service for both wired and wireless client networks.

Key concepts demonstrated include:

- The critical distinction between volatile running-config and persistent startup-config, and the consequence of not saving before a reload.
- Correct Layer 2 switch management using interface vlan 1 (SVI) instead of a physical port.
- VTY line Telnet configuration with password protection and enable secret for privileged access.
- DHCP pool configuration with excluded addresses for both wired (192.168.1.0/24) and wireless (192.168.2.0/24) subnets.
- Ping-based connectivity testing and physical fault simulation to illustrate real-world troubleshooting workflows.

Together, these labs build the end-to-end skills needed to set up, secure, and validate a small office LAN — skills directly relevant to the Cisco CCNA certification and real-world network administration roles.

C. GitHub README Description

Configure and Test Basic Network Connectivity

A hands-on Cisco Packet Tracer project covering the end-to-end configuration of a small office LAN. This project spans six lab videos and walks through switch setup, static and dynamic IP addressing, router interface activation, Layer 2 switch management via SVI, Telnet remote access, and dual-pool DHCP configuration for wired and wireless clients.

Topics Covered

- Cisco IOS CLI navigation and switch/router configuration
- Running-config vs. startup-config and configuration persistence
- Static IPv4 addressing (PCs, Laptop, Server, Router interface)
- VLAN 1 SVI configuration for Layer 2 switch management
- VTY Telnet access with password and enable secret
- DHCP pool configuration for wired and wireless subnets
- Ping-based connectivity testing and physical fault simulation

Technologies

Cisco Packet Tracer | Cisco IOS | Cisco 2960-24TT | Cisco ISR4331 | IPv4
| DHCP | Telnet | VLAN SVI

Relevance

Demonstrates practical ability to configure, secure, and verify a multi-device network from scratch.